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Image forming system having developer container, rotating member for rotating shutter, and restriction mechanism for rotating member

Abstract

An attached portion of an image forming apparatus includes a rotating member of which at least a part is exposed outside the image forming apparatus when the developer container is attached, and a rotation restricting mechanism that is movable between a restricting position to restrict rotation of the rotating member and an allowable position to allow rotation of the rotating member, and includes a restricting member that is at the restricting position when the developer container is not attached. The developer container is attached to the attached portion so that the discharging portion engaging portion and the shutter engaging portions are engaged with the frame engaged portions and the rotating member engaged portions respectively when the container shutter is in the close state, and the restricting member moves from the restricting position to the allowable position when the developer container is attached to the attached portion.

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Background/Summary

(1) This application is a continuation of application Ser. No. 17/979,815, filed Nov. 3, 2022, which is a continuation of application Ser. No. 17/558,762, filed Dec. 22, 2021, now U.S. Pat. No. 11,526,099, issued Dec. 13, 2022.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an image forming apparatus that forms an image on a recording material.

Description of the Related Art

(2) In an image electrophotographic type image forming apparatus, a configuration to replenish developer using a developer container, that is detachable from the image forming apparatus, is known. Japanese Patent Application Publication No. 2020-154300 discloses a configuration to open/close a shutter of a developer container for sealing an opening of the container attached to the image forming apparatus, by rotating the developer container.

SUMMARY OF THE INVENTION

(3) In a configuration where developer is replenished to an image forming apparatus using a developer container which is detachable from the image forming apparatus, it is an object of the present invention to provide a technique to improve operability of attaching the developer container to the image forming apparatus main body.

(4) To achieve this object, an image forming system of the present invention includes: a developer container; and an image forming apparatus including an attached portion to which the developer container is attached, wherein the developer container further comprises: a container portion that contains developer; a discharging portion in which a container opening through which the developer contained in the container portion is discharged to an outside of the developer container is provided, and which includes a discharging portion engaging portion; and a container shutter that is rotatable with respect to the discharging portion about a rotation axial line, between an open position in which the container shutter opens the container opening and a close position in which the container shutter closes the container opening, and that includes a shutter engaging portion; the attached portion further comprises: a frame in which a receiving port through which the frame receives the developer discharged from the container opening is provided, wherein the frame includes a frame engaged portion configured to engage with the discharging portion engaging portion so as to restrict a rotation of the discharging portion about the rotation axial line; a rotating member which is rotatable about the rotation axial line with respect to the frame and of which at least a part is exposed outside the image forming apparatus in a case where the developer container is attached to the attached portion, wherein the rotating member is provided outside the container shutter in a radius direction of a virtual circle centering the rotation axial line when viewed in a direction of the rotation axial line in a case where the developer container is attached to the

attached portion, and the rotating member includes a rotating member engaged portion that engages with the shutter engaging portion, and a restricted portion, and the rotating member is configured to rotate together with the container shutter by the rotating member engaged portion being engaged with the shutter engaging portion; and a rotation restricting mechanism that includes a restricting member having a restricting portion and movable between a restricting position in which the restricting portion restricts a rotation of the rotating member by engaging with the restricted portion of the rotating member and an allowable position in which the restricting portion allows the rotating member to rotate, and a restricting member is in the restricting position in a case where the developer container is not attached to the attached portion; wherein the developer container is attached to the attached portion in a state where the container shutter is at the close position so that the discharging portion engaging portion is engaged with the frame engaged portion and so that the shutter engaging portion is engaged with the rotating member engaged portion, wherein the restricting member is configured to move from the restricting position to the allowable position by an attachment of the developer container to the attached portion.

(5) According to the present invention, operability of attaching the developer container to the image forming apparatus main body can be improved.

(6) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an image forming apparatus;
- (2) FIG. 2 is a diagram for describing an internal configuration of the image forming apparatus;
- (3) FIG. 3 is a perspective view for describing a disposition of a circuit board;
- (4) FIG. 4 is a front perspective view for describing a disposition of the circuit board;
- (5) FIG. 5 is a perspective view of the circuit board and peripheral members thereof;
- (6) FIG. 6 is a side view of the circuit board and peripheral members thereof;
- (7) FIG. 7 is a top view of the circuit board and peripheral members thereof;
- (8) FIG. 8 is a perspective view for describing a configuration to hold a scanner unit and a drive motor;
- (9) FIG. 9 is a rear view of the circuit board in a direction vertical to the board surface;
- (10) FIG. 10 is a diagram for describing electronic components on the circuit board;
- (11) FIG. 11 is a block diagram for describing functions of the circuit board;
- (12) FIG. 12 is a side view for describing the positions of a replenishing portion and a scanner unit;
- (13) FIG. 13 is a top view for describing the positions of a replenishing portion and the scanner unit;
- (14) FIG. 14 is a perspective view of a developer container;
- (15) FIGS. 15A to 15C are enlarged perspective views of the replenishing portion;
- (16) FIGS. 16A to 16D are cross-sectional top views of the replenishing portion;
- (17) FIGS. 17A to 17C are exploded perspective views of a replenishing pack;
- (18) FIGS. 18A and 18B are diagrams for describing a configuration of the replenishing pack;
- (19) FIGS. 19A and 19B are exploded perspective views of an attached portion;
- (20) FIGS. 20A to 20C are perspective views of the attached portion;
- (21) FIGS. 21A and 21B are cross-sectional views of the replenishing pack and the attached portion;
- (22) FIG. 22 is a perspective view for describing a rotation locus of a mounting portion;
- (23) FIG. 23 is a top view for describing the rotation locus of the mounting portion;
- (24) FIG. 24 is a perspective view of a rotation restricting mechanism;

(25) FIGS. 25A and 25B are cross-sectional views when a lever lock mechanism is locked;
(26) FIGS. 26A and 26B are cross-sectional views when the lock of the lever lock mechanism is released;
(27) FIGS. 27A to 27E are cross-sectional views for describing a sequence of flow of the lever lock mechanism;
(28) FIGS. 28A and 28B are perspective views depicting a state when a discharge tray is opened/closed;
(29) FIGS. 29A and 29B are perspective views depicting a state when the replenishing pack is attached;
(30) FIG. 30 is a top view depicting a state where the replenishing pack is attached; and
(31) FIGS. 31A and 31B are perspective views of the replenishing portion and the developer container of which lever lock positions are different.

DESCRIPTION OF THE EMBODIMENTS

(32) Hereinafter, a description will be given, with reference to the drawings, of embodiments (examples) of the present invention. However, the sizes, materials, shapes, their relative arrangements, or the like of constituents described in the embodiments may be appropriately changed according to the configurations, various conditions, or the like of apparatuses to which the invention is applied. Therefore, the sizes, materials, shapes, their relative arrangements, or the like of the constituents described in the embodiments do not intend to limit the scope of the invention to the following embodiments.

Embodiment 1

(33) General Configuration of Image Forming Apparatus

(34) A general configuration of an image forming apparatus **1** according to an embodiment of the present invention will be described. The image forming apparatus **1** in of the present embodiment is a monochrome laser beam printer using an electrophotographic process, and forms an image on a recording material P using developer (toner), in accordance with image information sent from such an external device as a personal computer. Examples of the recording material P are: recording paper, label paper, OHP sheets and cloth.

(35) In the following description, the height direction of the image forming apparatus **1**, in a case where the image forming apparatus **1** is provided on a horizontal surface (upward vertical direction) is assumed to be the Z direction, and a direction that intersects with the Z direction and is parallel with the rotation axial line direction (main scanning direction) of a photosensitive drum **11** (described later) is assumed to be the X direction. A direction that intersects with the X direction and the Z direction is assumed to be the Y direction. The X direction, the Y direction and the Z direction preferably intersect so as to be orthogonal to each other. For convenience, the plus side of the X direction is called the “right side”, the minus side thereof is called the “left side”, the plus side of the Y direction is called the “front side” or “front surface side”, the minus side thereof is called the “rear side” or “rear surface side”, and the plus side of the Z direction is called the “upper side” and the minus side thereof is called the “lower side”.

(36) FIG. 1 is a perspective view of the image forming apparatus **1**, and FIG. 2 is a diagram for describing an internal configuration of the image forming apparatus **1** viewed in the X direction (rotation axial line direction of the photosensitive drum **11**). In FIG. 2, only the members related to the image forming process are selectively illustrated. In FIG. 1, the image forming apparatus **1** includes a feeding cassette **4** in which the recording material P is stored, and a discharge tray **14** on which the discharged recording material P is loaded. The feeding cassette **4** can be pulled out in the Y direction so that the user can replenish the recording material P. An image is formed on the recording material P, which was fed from the feeding cassette **4**, and the recording material P is discharged from a discharge port **15** in the discharge direction (Y direction), as indicated in FIG. 1, and is loaded on the discharge tray **14**. A front cover **70** is provided on a part of the end surface (a part of the front surface) of the image forming apparatus **1** on the downstream side in the discharge

direction, so as to cover a later mentioned circuit board **100**. An exterior cover **71** is provided on the front surface (on a part other than the area where the front cover **70** is provided), and on a side surface and a top surface of the image forming apparatus **1**. The front cover **70**, the exterior cover **71** and the above mentioned discharge tray **14** constitute a housing **720** of the image forming apparatus **1**. The housing **720** is a member that covers the entire image forming apparatus **1**, and includes the later mentioned scanner unit **50** and other process members. The above mentioned discharge outlet **15** is an opening that is formed in a part of the housing **720**, and the recording material P is discharged out of the image forming apparatus **1** via the discharge port **15**.

(37) A flow of the image forming operation performed on the recording material P will be described with reference to FIG. **2**. When image information is sent to the image forming apparatus **1**, the photosensitive drum **11** is rotary-driven in an arrow R direction at a predetermined peripheral velocity (process speed). The scanner unit **50** emits a laser beam onto the photosensitive drum **11** based on the inputted image information. The scanner unit **50** is a unit configured of: a laser oscillator that outputs the laser beam; a polygon mirror and lenses through which the laser beam is emitted onto the photosensitive drum **11**; a scanner motor to rotate the polygon mirror; and a frame that integrally supports these members. The photosensitive drum **11** is charged in advance by a charging roller **17**, and an electrostatic latent image is formed on the photosensitive drum **11** (image bearing member) by the irradiation of the laser beam. Then toner (developer) contained in a container portion **18** is carried to the photosensitive drum **11** (photosensitive member) by a developing roller **12** (developer bearing member), whereby the electrostatic latent image is developed, and a toner image (developer image) is formed on the photosensitive drum **11**. To the container portion **18**, a later mentioned attached portion **200** for forming a replenishing port, to replenish toner from the outside, is connected.

(38) In parallel with the above mentioned image forming process, the recording material P is fed from the feeding cassette **4**. On the conveying path of the image forming apparatus **1**, a pickup roller **3**, a feeding roller **5a**, and a conveying roller pair **5c** are provided. The pickup roller **3** contacts with the topmost recording material P out of the recording materials P loaded in the feeding cassette **4**, and feeds the recording material P by the rotation of the roller. The feeding roller **5a** and a separation roller **5b**, that press-contacts thereto, form a separation nip. In a case where a plurality of recording materials P are fed to the separation nip due to the influence of friction force between the recording materials P, the feeding roller **5a** and the separation roller **5b** separate the plurality of recording materials P so that only the topmost recording material P is fed to the downstream side.

(39) The recording material P fed from the feeding cassette **4** is conveyed toward a transfer roller **7** via the conveying roller pair **5c**. By the transfer bias applied to the transfer roller **7**, the toner image, formed on the photosensitive drum **11**, is transferred to the recording material P. The recording material P, on which the toner image was transferred by the transfer roller **7**, is heated and pressed by a fixing apparatus **9**, whereby the toner image is fixed to the recording material P. The fixing apparatus **9** is constituted of a heat roller **9a** which includes a fixing heater **9c**, and a pressure roller **9b** which is biased toward the heat roller **9a**. Then the recording material P, on which the toner image was fixed, is discharged to the discharge tray **14** via a discharge roller pair **10**. In the case of forming an image on both sides of the recording material P, the discharge roller pair **10** switches back (changes the conveying direction of) the recording material P (on which first surface an image was formed), so as to guide the recording material P to a double-sided conveying path **16**. The recording material P, guided to the double-sided conveying path **16**, is conveyed toward the transfer roller **7** again via a double-sided conveying roller pair **5d**. After an image is formed on the second surface of the recording material P by the transfer roller **7**, the recording material P is discharged to the outside via the discharge roller pair **10**. The toner, which remains on the photosensitive drum **11** after the toner image is transferred to the recording material P, is cleaned by a cleaning unit **13**.

(40) As illustrated in FIG. 2, the image forming apparatus **1** includes the circuit board **100**. The circuit board **100** is constituted of a wiring board **101** formed by insulating material, and electronic components **111** and **121** which are soldered to the wiring board **101**. The electronic components **111** and **121** are electrically connected since conductor wires are formed on and inside the board of the wiring board **101**. The circuit board **100** has a function to convert AC current supplied from outside the image forming apparatus **1** into DC current, and to convert input voltage so as to acquire a predetermined voltage value required for the image forming process. As illustrated in FIG. 2, the circuit board **100** is provided so that the surface of the wiring board **101**, on which the electronic components **111** and **121** are mounted, intersects with the discharge direction. Further, the wiring board **101** is provided between the front cover **70** and the scanner unit **50** in the discharge direction. The electronic components **111** and **121** are provided on the wiring board **101** on the side facing the scanner unit **50**.

(41) Disposition of Circuit Board

(42) A disposition of the circuit board **100** according to the present embodiment will be described in detail with reference to FIGS. 3 to 8. FIG. 3 is a perspective view of the image forming apparatus **1** for describing a disposition of the circuit board **100**, and unlike FIG. 1, illustration of the front cover **70** and the exterior cover **71** are omitted. In FIG. 3, an attached portion **200** to replenish the toner is illustrated anew. In the image forming apparatus **1** of the present embodiment, the user or service personnel can replenish developer via the attached portion **200**, and the attached portion **200** is connected to the container portion **18** inside the image forming apparatus **1**. The attached portion **200** will be described in detail later.

(43) As illustrated in FIG. 3, the circuit board **100** is provided on the front surface side, and the scanner unit **50** and a drive motor **60** (drive source) are provided on the rear side (minus side of the Y direction) of the circuit board **100**. In FIG. 3, the scanner unit **50** and the drive motor **60** are at positions which cannot be seen, hence are indicated by dotted lines. As illustrated in FIG. 3, the image forming apparatus **1** includes a right side plate frame **72**, a left side plate frame **73**, and a base frame **74**. The circuit board **100** is supported by these frame members, and is equipped in the image forming apparatus **1** so that the board surface of the circuit board **100** is approximately parallel with the XZ plane. In each side edge portion of the right side plate frame **72** and the left side plate frame **73**, bent portions **72a** and **73a** are formed respectively. The bent portion **72a** is bent toward the plus side of the X direction so as to be approximately parallel with the XZ plane, and the bent portion **73a** is bent toward the minus side of the X direction so as to be approximately parallel with the XZ plane. By bending both of the side plate frames toward the outside of the image forming apparatus **1** in this way, electronic components can be mounted on a larger region of the wiring board **101**.

(44) FIG. 4 is a front perspective view of the image forming apparatus **1** for describing a disposition of the circuit board **100**. As illustrated in FIG. 4, a distance L1 between an inner surface of the right side plate frame **72** and that of the left side plate frame **73** in the X direction is shorter than a length L2 of the circuit board **100** in the X direction. The wiring board **101** is provided on the plus side (front surface side) of the bent portions **72a** and **73a** in the Y direction. The circuit board **100** and the bent portions **72a** and **73a** overlap in the front view. In FIG. 4, the bent portions **72a** and **73a**, the scanner unit **50** and a part of the drive motor **60** are at positions which cannot be seen, hence are indicated by dotted lines.

(45) As described above, the circuit board **100** is provided on the front surface side, and is formed throughout the area from the right side plate frame **72** to the left side plate frame **73**, hence in the image forming apparatus **1**, there is no need to provide bundle wire, or the like, that crosses between the right side plate frame **72** and the left side plate frame **73** in the X direction. Thereby the bundle wire length can be shorter than prior arts, and cost can be reduced accordingly. Further, the region where the bundle wire is routed is also less than prior arts, which means that electric noise can be reduced as well.

(46) Positional Relationship Between Electronic Components and Scanner Unit

(47) A positional relationship between the electronic components **111** and a scanner unit **50** will be described in detail with reference to FIGS. 5 to 7. FIG. 5 is a perspective view of the circuit board **100** viewed from the rear side of the main body. The electronic components **111**, of which sizes in the Y direction are larger than the other members, are located at the lower part of the wiring board **101**, and are mounted so as to be contained below the scanner unit **50**, so as to utilize the space efficiently. A power supply input portion **115** is provided at the edge of the wiring board **101**. The power supply input portion **115** is connected to a later mentioned inlet **116**, so that power is supplied from a commercial power supply.

(48) FIG. 6 is a diagram depicting the circuit board **100** viewed from the left side surface of the main body. A part of the scanner unit **50** is at a position overlapping with the attached portion **200**, and cannot be seen, hence is indicated by a dashed line. The scanner unit **50** is provided at an optimum position to emit a laser beam, indicated by a dotted line, to the photosensitive drum **11**. In an area where the scanner unit **50** and the wiring board **101** are closest to each other in the Y direction, no members which protrude significantly from the board surface, such as the electronic components **111**, are provided. In other words, the scanner unit **50** and the electronic components **111** are provided shifted from each other in the Z direction, so as not to interfere with each other.

(49) FIG. 7 is an enlarged top view when the circuit board **100** is viewed from the top surface of the main body. As illustrated in FIG. 7, the scanner unit **50** and the electronic components **111** are provided at positions that partially overlap with each other. The scanner unit **50** is on the upper side of the electronic components **111**, as mentioned above, which means that the electronic components **111** cannot be seen in this direction. Therefore in FIG. 7, in order to clearly indicate the positional relationship of these two members, the scanner unit **50** is indicated by a dotted line, as if the electronic components **111** were transparent. By disposing the electronic components **111** in this position, the distance between the circuit board **100** and the scanner unit **50** in the Y direction (front—back direction) can be shortened, and the image forming apparatus **1** can be downsized.

(50) Positional Relationship Between Electronic Component and Drive Motor

(51) A positional relationship between the electronic components **111** and the drive motor **60** will be described in detail with reference to FIGS. 5 to 7. The drive motor **60** has a function to rotate the conveying members (e.g. pickup roller **3**, feeding roller **5a**, conveying roller pair **5c**) and the photosensitive drum **11**, to feed and convey the recording material P.

(52) As illustrated in FIG. 5, the drive motor **60** projects to the minus side of the X direction, and the wiring board **101** is provided on the front side of the main body with respect to the drive motor **60**. The electronic components **111** are mounted distant from the drive motor **60**, so as not to interfere with the drive motor **60**. As illustrated in FIG. 6, the drive motor **60** and the electronic components **111** are provided at positions that partially overlap with each other when viewed from the left side surface of the main body. Further, as illustrated in FIG. 7, the drive motor **60** and the electronic components **111** are shifted from each other in the X direction when viewed from the top surface of the main body, so as not to interfere with each other. By disposing the electronic components **111** in these positions, the distance between the circuit board **100** and the drive motor **60** in the Y direction (front-back direction) can be shortened, and the image forming apparatus **1** can be downsized.

(53) Configuration of Mounting to Main Body

(54) A configuration of mounting the scanner unit **50** and the drive motor **60** to the main body will be described in detail with reference to FIG. 8. FIG. 8 is a perspective view in FIG. 5, to which the right side plate frame **72** and a scanner holding member **40** are added. Illustration of the left side plate frame **73** and the base frame **74** is omitted here. The scanner unit **50** is held by the scanner holding member **40**. The scanner holding member **40** is fixed to the right side plate frame **72** and the left side plate frame **73** (not illustrated in FIG. 8) respectively, and is provided below the attached portion **200** so as to bridge these two frames. The drive motor **60**, on the other hand, is

mounted on the right side plate frame **72**, and a gear, connected to the drive motor **60**, is provided on the plus side (right side) of the right side plate frame **72** in the X direction. The drive force of the drive motor **60** is transferred to the feeding roller **5a** and the photosensitive drum **11** via this gear.

(55) Configuration of Circuit Board

(56) A configuration of the circuit board **100** will be described with reference to FIGS. **9** and **10**. FIG. **9** is a rear view of the circuit board **100** viewed from the rear side of the main body. In FIG. **9**, not only the circuit board **100** but also the scanner unit **50**, the drive motor **60**, and the attached portion **200** are illustrated. In FIG. **10**, only the circuit board **100** is illustrated. The circuit board **100** is constituted of the low voltage power supply portion **110** which receives AC power from an external commercial power supply, and converts the AC power into DC power, and a high voltage power supply portion **120** which supplies high voltage required for image formation to each process member. In the circuit board **100** of the present embodiment, the low voltage power supply portion **110** and the high voltage power supply portion **120** are mounted on the same board. The low voltage power supply portion **110** includes: a low voltage power transformer **112**, a heat sink **113**, and an electrolytic capacitor **114** as the electronic components **111** of which sizes in the Y direction are large. The low voltage power supply portion **110** also includes the power supply input portion **115**. The high voltage power supply portion **120** includes a charging transformer **122**, a developing transformer **123** and a transferring transformer **124** as the electronic components **121** of which sizes in the Y direction are large. As illustrated in FIG. **9**, the electronic components **111** and **121**, of which sizes in the Y direction are large, are all provided distant from the positions of the scanner unit **50**, the drive motor **60** and the attached portion **200**.

(57) Other components provided on the circuit board **100** will be described with reference to FIG. **10**. On the upper and lower edges of the circuit board **100**, a plurality of connectors **220**, **221**, **222** and **223** are provided, and the circuit board **100** is connected with various members by bundle wires. The connector **220** is connected to a drive motor **60** and sensors (not illustrated) to detect the recording material P in-conveying, for example. The connector **221** is connected to a laser output portion (not illustrated) of the scanner unit **50** and a scanner motor (not illustrated) to rotate a polygon mirror. The connector **222** is connected to a control panel (not illustrated) which includes a power switch, an execution key, and the like, operated by the user, and a video controller **140**. The connector **223** is connected to the fixing heater **9c**. On a shaded portion **224** facing the drive motor **60**, electronic components, of which sizes in the Y direction are small in the high voltage power supply portion **120**, are mounted. Specifically, resistors, jumper wires, and the like, are provided. Resistors provided at this position have a function to adjust various biases outputted from the charging transformer **122**, the developing transformer **123**, and the transferring transformer **124**.

(58) The functions of the low voltage power supply portion **110** and the high voltage power supply portion **120** will be described with reference to FIGS. **9** and **11**. FIG. **11** is a block diagram for describing the functions of the circuit board **100**. First, the low voltage power supply portion **110** receives power from an external power supply via the power supply input portion **115** mounted on the edge of the board, and converts the AC voltage into stable DC voltage by a rectifying/smoothing circuit which includes the electrolytic capacitor **114**. Then the low voltage power supply portion **110** converts the DC voltage into high frequency AC voltage using such a switching element as a transistor, and inputs the high frequency AC voltage to the low voltage power transformer **112**. The low voltage power transformer **112** converts the high frequency AC voltage (input voltage) into AC voltage having a desired voltage value (output voltage). The low voltage power supply portion **110** converts the AC voltage into the DC voltage again, and outputs the acquired DC voltage to the high voltage power supply portion **120**. In the low voltage power supply portion **110**, energy loss of an individual circuit component generates heat, hence the heat sink **113**, made of aluminum or iron, is provided to release the heat.

(59) The high voltage power supply portion **120** converts the voltage supplied from the low voltage

power supply portion **110** (e.g. 24 V) into a high voltage that is required for the image forming process, which includes charging, developing and transferring. The charging transformer **122** converts the voltage supplied from the low voltage power supply portion **110** into voltage for charging, and the converted voltage is then supplied to the charging roller **17**. The developing transformer **123** converts the voltage supplied from the low voltage power supply portion **110** into a voltage for developing, and the converted voltage is then supplied to the developing roller **12**. The transferring transformer **124** converts the voltage supplied from the low voltage power supply portion **110** into a voltage for transferring, and the converted voltage is then supplied to the transfer roller **7**.

(60) The low voltage power supply portion **110** supplies voltage (e.g. 3.3 V or 5 V) not only to the high voltage power supply portion **120**, but also to the scanner unit **50**, the drive motor **60**, an engine controller **130** and a video controller **140**. Here, the engine controller **130** has a function to integrally control various process members. The engine controller **130** includes a CPU (not illustrated), a RAM (not illustrated) that calculates data required for controlling the image forming apparatus **1** and temporarily stores data, and a ROM (not illustrated) that stores programs and various data to control the image forming apparatus **1**. The video controller **140** has a function to receive print data by communicating with such an external device as a personal computer, and to notify a result of analyzing the print data to the engine controller **130**. The engine controller **130** and the video controller **140** may be provided on a board that is different from the circuit board **100**, or may be provided on the same board.

(61) The AC power, which the power supply input portion **115** received from a commercial power supply, is supplied not only to the low voltage power supply portion **110**, but also to the fixing heater **9c**. In the circuit board **100** illustrated in FIG. **10**, a triac (not illustrated) is provided between the power supply input portion **115** and the connector **223**, and the temperature of the fixing heater **9c** can be adjusted by switching the triac ON/OFF, so as to change the sine wave. Driving of the roller and the like in the fixing apparatus **9** is performed by the drive motor **60**.

(62) Disposition and Configuration of Replenishing Unit

(63) The attached portion **200** will be described with reference to FIGS. **12** to **22**. As mentioned above, in the image forming apparatus **1**, the attached portion **200**, which replenishes toner from an outside source when the residual amount of toner in the container portion **18** is low, is provided so that the container portion **18** need not be detached from the housing **720**. In the attached portion **200**, which is a toner container (developer container), a replenishing pack **210** can be detached/attached.

(64) FIG. **12** is a left side view of the image forming apparatus **1** viewed in the rotation axial line direction of the photosensitive drum **11**. In FIG. **12**, the exterior cover **71** and the left side plate frame **73** have been removed. The attached portion **200** is constituted of: a mounting portion **201** where the replenishing pack **210** (not illustrated in FIG. **12**) is mounted; a toner-receiving portion **202** which is a cylindrical-shaped frame; and a replenishing path portion **203** that connects the container portion **18** and the toner-receiving portion **202**. The mounting portion **201** includes a replenishing port **204**, which is an opening to replenish toner, and is configured such that the toner which passed through the replenishing port **204** moves on to the toner-receiving portion **202** and the replenishing path portion **203** sequentially, and is finally supplied to the container portion **18**. A part of the scanner unit **50** is at a position which overlaps with the attached portion **200**, and cannot be seen, hence this region is indicated by a dotted line in FIG. **12**. Specifically, the toner-receiving portion **202** and the replenishing path portion **203** of the attached portion **200** overlap with the scanner unit **50**. In other words, the toner-receiving portion **202** and the replenishing path portion **203** are at positions overlapping with the scanner unit **50** in the Z direction. Here, if it is assumed that R1 is a region where the replenishing port **204** is provided in the Y direction (horizontal direction), and R2 is a region where the scanner unit **50** is provided in the Y direction, then R1 and R2 overlap.

(65) It is also assumed that S is a virtual plane that passes through an upper edge **18b**, which is located at the uppermost side of a frame **18a** of the container portion **18**, and is parallel with the horizontal plane. The virtual plane S is indicated by dashed lines in FIG. **12**. If the virtual plane S is a reference, then a part of the attached portion **200** is located at the plus side (upper side) in the Z direction. In other words, a part of the attached portion **200** projects upward from the upper edge **18b** of the container portion **18**. Specifically, a part of the attached portion **200** includes the entire mounting portion **201**, a part of the toner-receiving portion **202**, and a part of the replenishing path portion **203**. The part of the toner-receiving portion **202** and the part of the replenishing path portion **203**, which project upward from the virtual plane S, overlap with the scanner unit **50**.

(66) As illustrated in FIG. **12**, a part of the container portion **18** is at a position overlapping with a drum frame **11a** that supports the photosensitive drum **11**, and cannot be seen, hence this region is indicated by a dotted line. The container portion **18** supports the developing roller **12** which bears the developer, and this developing roller **12** is also at a position that cannot be seen, hence is indicated by a dotted line in FIG. **12**.

(67) FIG. **13** is a top view of the image forming apparatus **1** when the exterior cover **71** is removed. As mentioned above, the mounting portion **201** includes the replenishing port **204**. The mounting portion **201** also includes a ring portion **201a** which is provided to surround the replenishing port **204**, and a lever portion **201b** which is connected to the ring portion **201a**. As illustrated in FIG. **13**, the width of the attached portion **200** in the X direction is shorter than the width of the container portion **18** in the X direction.

(68) Here, the laser beam emitted from the scanner unit **50** onto the photosensitive drum **11** spreads in a trapezoidal shape, as illustrated in FIG. **13**, due to the functions of the polygon mirror and the lens (neither illustrated). Therefore the width of the scanner unit **50** in the X direction is shorter than the width of the photosensitive drum **11**. As a result, a space is created between the left end of the scanner unit **50** and the left side plate frame **73**, and in the present embodiment, the attached portion **200** is provided in this space. In other words, the attached portion **200** is provided between the scanner unit **50** and the left side plate frame **73** in the X direction, as illustrated in FIG. **13**. Further, the replenishing port **204** and the scanner unit **50** are provided side-by-side in the X direction, in a region where the container portion **18** is provided. If the attached portion **200** is provided at this position, the size of the image forming apparatus **1** is little influenced by disposing the attached portion **200**.

(69) The attached portion **200** is provided on opposite sides of the drive motor **60** with respect to the scanner unit **50**. The drive motor **60** used in the present embodiment is relatively small, hence as illustrated in FIG. **12**, the attached portion **200** and the drive motor **60** do not overlap in the Z direction. Therefore it is possible to provide the attached portion **200** and the drive motor **60** on the same side with respect to the scanner unit **50**, but in the case of using a larger drive motor **60**, the attached portion **200** must be provided at an upper position, which increases the size of the image forming apparatus **1**. If the configuration where the attached portion **200** is provided on the opposite side of the drive motor **60** is used, as described in the present embodiment, a larger drive motor **60** can be used without increasing the size of the image forming apparatus **1**. In other words, design flexibility can be improved.

(70) Description of Replenishing Configuration

(71) FIG. **14** is a perspective view of a developer container **230** constituted of the container portion **18** and the attached portion **200**. In FIG. **14**, illustration of a part of members associated with the attached portion **200** is omitted. On the inner wall of the cylindrical-shaped toner-receiving portion **202**, a side surface opening **205**, which is connected to the replenishing path portion **203**, is formed, but the side surface opening **205** is covered by a main body shutter portion **206** and cannot be seen, hence is indicated by dotted lines. Toner is guided from the toner-receiving portion **202** to the replenishing path portion **203** via this side surface opening **205**, and is then contained in the container portion **18** via the replenishing path portion **203**. On the main body shutter portion **206**, a

main body shutter portion driving transfer rib **206a** is provided and used to receive driving force for the replenishing pack **210**, and to rotate the main body shutter portion **206** (details described later).

(72) FIGS. **15A** to **15C** are enlarged perspective views of the attached portion **200**, and FIGS. **16A** to **16D** are enlarged cross-sectional views of the mounting portion **201** viewed from the top surface. In FIG. **15A**, the side surface opening **205** formed in the toner-receiving portion **202** is covered by the main body shutter portion **206**, and cannot be seen, hence is indicated by a dotted line. The main body shutter portion **206** is a cylindrical-shaped member which is concentric with the toner-receiving portion **202**, and is provided at the inside of the toner-receiving portion **202**. An opening **207** for the toner to pass through is also formed in the main body shutter portion **206**. In FIG. **15A**, the side surface opening **205** and the opening **207** are provided at positions that are shifted from each other, hence the side surface opening **205** is closed. The mounting portion **201** is provided in a top surface portion **240** (not illustrated), so as to be concentric with the toner-receiving portion **202**, and is restricted in the rotating direction by a later mentioned lever lock mechanism when the replenishing pack **210** is not attached. This lever mechanism can prevent a shift in phases of a mounting portion driving transfer rib **201d** and the main body shutter portion driving transfer rib **206a** before attaching the replenishing pack **210**, so that the replenishing pack **210** can be attached with certainty.

(73) As illustrated in FIGS. **15B** and **16C**, when the replenishing pack **210** is attached, a lever lock release rib **214a**, which is a release portion provided in pack shutter portion **214** (shutter member), is inserted into a lock release recessed portion **201f**. In FIGS. **15B** and **16C**, only the pack shutter portion **214** of the replenishing pack **210** is illustrated. The lever lock release rib **214a** is a projecting portion that projects from an outer surface of the pack shutter portion **214**, which extends in the rotation axial line direction, in a direction intersecting with the rotation axial line. When the replenishing pack **210** is attached to the attached portion **200**, the lever lock release rib **214a** moves a lever lock release link **229**, whereby the lever lock mechanism is released, and the lever portion **201b** can be rotated. The lever lock mechanism will be described in detail later. A state of engagement between a guided portion **232** of an inserting portion **212** (discharging portion engaging portion) and a guiding portion **247** (**248**) of the toner-receiving portion **202** (frame engaged portion) is as illustrated in FIG. **16B**. By this engaging configuration, the relative movement between the inserting portion **212** and the toner-receiving portion **202**, in the circumferential direction centering the rotation axial line of the pack shutter portion **214**, can be mutually restricted. Details will be described later.

(74) The mounting portion driving transfer rib **201d** is engaged with the main body shutter portion driving transfer rib **206a** via a pair of driving transfer surfaces **214b** of the pack shutter portion **214** of the replenishing pack **210**. The mounting portion driving transfer rib **201d** is a protruded portion (rotting member protruded portion) that extends from an inner peripheral surface, centering the rotation axial line R of a hole formed in the approximately annular mounting portion **201**, to the rotation axial line RA. The mounting portion driving transfer rib **201d** includes a first engaging portion **201d1** and a second engaging portion **201d2**, which are portions to form a first projected portion side surface **201ds1** and a second projected portion side surface **201ds2**, and which are arranged across a projected portion front end surface **201dt** in the circumferential direction centering the rotation axial line RA. The main body shutter portion driving transfer rib **206a** is a projected portion (shutter convex portion) that extends from the inner peripheral surface, centering the rotation axial line R of the hole formed in the approximately cylindrical main body shutter portion **206**, to the rotation axial line RA. The main body shutter portion driving transfer rib **206a** includes a first shutter projected portion side surface **206a1** and a second shutter projected portion side surface **206a2**, which are arranged across a shutter projected portion front end surface **206at** in the circumferential direction centering the rotation axial line RA. The main body shutter portion **206** can be rotated inside the toner-receiving portion **202** by the user holding the lever portion **201b**

and moving the lever portion **201b** from the state in FIGS. **15B** and **16C** to the state in FIGS. **15C** and **16D**. In FIGS. **15C** and **16D**, the side surface opening **205** and the opening **207** are at overlapping positions, hence the side surface opening **205** is opened, and toner can be replenished through the side surface opening **205**.

(75) In the case where an image is formed on the recording material **P**, the side surface opening **205** must be shut off (closed) so that the toner being stirred in the container portion **18** by the stirring member (not illustrated) does not leak through the side surface opening **205**. Hence, when an image is formed, the lever portion **201b** is moved to the position indicated in FIGS. **15A** and **16A**. This position is referred to as an initial position or operation position of the lever portion **201b**. In the case where toner is replenished from the replenishing pack **210** (described later) to the container portion **18**, on the other hand, the side surface opening **205** must be opened. Hence, when toner is replenished, the lever portion **201b** is moved to the position indicated in FIGS. **15C** and **16D**. This position is referred to as a replenishing position of the lever portion **201b**.

(76) Here, it is preferable that the size of the lever portion **201b** is as large as possible, so that the user can hold the lever portion **201b** easily. The configuration of the operation portion for the user to open/close the shutter is not limited to the configuration using the lever portion **201b**. Various configurations may be used as long as at least a part of the mounting portion **201** is exposed in a state where the replenishing pack **210** is mounted so that the user can operate the part. Further, the circuit board **100** is provided on the front side (plus side of Y direction) of the attached portion **200**, and in the present embodiment, the attached portion **200** and the circuit board **100** are provided in proximity to each other in order to downsize the image forming apparatus **1** in the Y direction.

Therefore, as illustrated in FIGS. **22** and **23**, a notch **101a** is formed on the upper edge of the wiring board **101**, so as to prevent contact with the lever portion **201b**. FIG. **22** is a perspective view viewed from the rear side of the main body, and FIG. **23** is a top view. In FIG. **23** the position corresponding to the notch **101a** is indicated by a dotted line. The lever portion **201b** in the initial position overlaps with the wiring board **101**. As illustrated in FIGS. **22** and **23**, the notch **101a** is formed at a position corresponding to the rotation locus of the lever portion **201b**. In the present embodiment, the notch **101a** is formed on the wiring board **101**, but interference with the lever portion **201b** may be prevented by forming a through hole or a groove in the wiring board **101**.

(77) The configuration of receiving replenishment of the developer in the image forming apparatus or the image forming system according to the present embodiment will be described in detail with reference to FIGS. **19A**, **19B**, **20A** to **20C**, **21A**, and **21B**.

(78) FIGS. **19A** and **19B** are exploded perspective view of the attached portion **200**. FIGS. **19A** and **19B** are perspective views viewed from different directions. FIGS. **20A** to **20C** are perspective views of the attached portion **200**, and unlike FIGS. **15A** to **15C**, illustration of the replenishing pack **210**, a lever lock member **227** and a lever lock portion pressing spring **228** is omitted. FIGS. **20A** and **20C** indicate a state where the mounting portion **201** (opening/closing member) is at a closed function position viewed from different perspective directions. The mounting portion **201** is provided so as to be located outside the pack shutter portion **214** in the radius direction r of a virtual circle **VC**, centering the rotation axial line **RA**, when the mounting portion **201** is viewed in the rotation axial line direction in the state where the replenishing pack **210** is attached. FIG. **20B** indicates a state where the mounting portion **201** is at an open function position. The main body shutter portion **206** (apparatus shutter) is provided on the inside of the toner-receiving portion **202** (developer-receiving portion). The main body shutter portion **206** includes the opening **207** (apparatus opening) to form a communication hole to replenish toner, and a sealing member **243** is adhered to the inner peripheral surface of the main body shutter portion **206**, so as to surround the outer periphery of the opening **207**. When the main body shutter portion **206** is at the open position, the sealing member **243** contacts with the outer surface of the inserting portion **212**, particularly around the opening **213**, and seals the later mentioned communication hole **245**. On the inside of the communication hole **245**, guiding portions **247** and **248** (frame engaged portions) that guide a

guided portion **232** (discharging portion engaging portion) of the inserting portion **212** of the replenishing pack **210** are provided. Furthermore, the main body shutter portion driving transfer rib **206a**, which guides a positioning portion **217** (a guided portion of the pack shutter portion **214** (cylindrical container shutter) is also provided on the inner peripheral surface of the main body shutter portion **206**. At the opening side of a recessed inserting portion configuration of the attached portion **200** where a protruded inserting portion **212** of the replenishing pack **210** is inserted, a mounting portion **201** (opening/closing member) is rotatably mounted. The mounting portion driving transfer rib **201d**, which projects inward from the inner peripheral surface side of the mounting portion **201**, is also a guiding portion that guides the positioning portion **217** of the pack shutter portion **214**. When the pack shutter portion **214** is at the close position, the guided portion **232** can be engaged with the guiding portions **247** and **248**, and the positioning portion **217** can be engaged with the mounting portion driving transfer rib **201d** and the main body shutter portion driving transfer rib **206a**.

(79) FIGS. **21A** and **21B** are schematic cross-sectional views of the replenishing pack **210** and the attached portion **200**. FIG. **21A** indicates a state where the mounting portion **201** is at the close function position, and each shutter is at the close position. In other words, the opening **213** (container opening) is closed by the sealing member **231** provided on the inner peripheral surface of the pack shutter **214**, and the side surface opening **205** (receiving port) is closed by the main body shutter portion **206**. FIG. **21B** indicates a state where the mounting portion **201** is at the open function position, and each shutter is at the open position. In other words, the opening **213** and the receiving port **205** communicate through a communication hole **245**, which is formed by the opening **207** provided in the main body shutter portion **206**, and the opening **244** of the sealing member **243**, provided on the inner peripheral surface of the main body shutter portion **206** so as to surround the opening **207**. In other words, this is a state where toner (developer) can be replenished from the replenishing pack **210** (developer container) to the image forming apparatus **1**.

(80) Configuration of Replenishing Container

(81) Configuration of the replenishing pack **210** (replenishing container) will be described with reference to FIGS. **18A** and **18B**. The replenishing pack **210** includes: a pouch portion **211** containing toner to be replenished; a discharging portion (nozzle portion, pipe, tube, valve) **212** that is a cylindrical-shaped inserting portion to be inserted into the replenishing port **204** that discharges the toner inside the pouch portion; an opening **213** that is formed on the side surface of the inserting portion **212** for the toner to pass through; and the pack shutter portion **214** that covers the opening **213** to prevent the leakage of toner. The pouch portion **211** is formed to be flatter in the direction toward the opposite side of the inserting portion **212**, and a pouch edge **216**, extending in a predetermined direction, is formed on the edge. The pouch portion **211** is a pouch formed by performing pouch processing on a flexible polypropylene sheet, but the container portion of the toner is not limited to a pouch, and may be a resin bottle, a paper or vinyl container, or the like.

(82) As illustrated in FIG. **18A**, a pack shutter portion opening **214c** is formed on the pack shutter portion **214**. When the pack shutter portion **214** is rotated and the pack shutter portion opening **214c** and the opening **213** of the inserting portion **212** match (rotation phase positions thereof overlap), toner can be replenished from the replenishing pack **210**. FIG. **18B** is a view of the replenishing pack **210** viewed from an angle different from FIG. **18A**. The opening **213** formed in the inserting portion **212** is covered by the pack shutter portion **214**, and cannot be seen, hence is indicated by a dotted line.

(83) The pack shutter portion **214** is a cylindrical-shaped member that is concentric with the inserting portion **212**, and is rotatably provided at the outer side (at outer periphery) of the inserting portion **212**. On the outer peripheral surface of the pack shutter portion **214**, a positioning portion **217**, which engages with the mounting portion **201** and the driving transfer surfaces **214b**, which face each other via the positioning portion **217** in the circumferential direction of the outer periphery of the pack shutter portion **214**, are provided. In other words, on the outer peripheral

surface of the pack shutter portion **214**, a groove (a recessed portion that is recessed inward from the outer peripheral surface of the pack shutter portion **214** in the radius direction r of a virtual circle VC), which is constituted of the positioning portion **217** as a groove bottom surface (recessed portion bottom surface) and the driving transfer surfaces **214b** as groove side surfaces, is formed. This groove is open at the front end portion of the outer peripheral surface of the pack shutter portion **214** in the inserting direction of the inserting portion **212**. The pack shutter portion **214** is rotated with respect to the inserting portion **212** by the driving transfer surfaces **214b** receiving force in the circumferential direction from the mounting portion driving transfer rib **201d**. Between the driving transfer surfaces **214b**, a rib **214a**, to release the lever lock mechanism of the mounting portion **201**, is provided. By disposing the rib **214a** to release the lever lock mechanism between the driving transfer surfaces **214b**, the lever lock mechanism of the mounting portion **201** can be accurately released when the replenishing pack **210** is attached.

(84) A configuration of the replenishing pack (developer container) used for the image forming apparatus or the developer replenishing system according to the present embodiment will be described in detail with reference to FIGS. **17A** to **17C**, **18A**, and **18B**. FIGS. **17A** to **17C** are exploded perspective views of the replenishing pack **210**. FIGS. **17A** to **17C** are perspective views viewed from different directions. In the inserting portion **212**, when the pack shutter portion **214** is at a close position, a pack shutter portion opening **214c** provided in the pack shutter portion **214**, and the guided portion **232** provided so as to be recessed from the outer peripheral surface of the inserting portion **212**, overlap with each other in the rotation phase in the circumferential direction. In this state, the replenishing pack **210** is in a state where the guided portion **232** is engaged with and guided by the guiding portions **247** and **248**, and the pack shutter portion opening **214c** is engaged with the periphery of the sealing member **243** provided on the inner peripheral surface of the main body shutter portion **206**. In the state where the replenishing pack **210** is attached to the attached portion **200**, a first guided portion **232a** (on the upstream side in the inserting direction) of the guided portion **232** engages with the guiding portion **247**, and a second guided portion **232b** (on the downstream side in the inserting direction) engages with the guiding portion **248**. Further, a surface (step difference portion) extending in the circumferential direction between the first guided portion **232a** and the second guided portion **232b**, engages in the inserting direction with a surface (step difference portion) between the guiding portion **247** and the guiding portion **248**, extending in the circumferential direction, and determines a position in the inserting direction between the inserting portion **212** and the mounting portion **201**. The pack shutter portion opening **214c** is a notch-shaped opening, of which width increases in a direction toward the front end of the inserting portion **212**. A pair of facing portions, which form the pack shutter portion opening **214c** and face each other in the circumferential direction, are provided on each side of the sealing member **243** in the circumferential direction.

(85) As the shutter engaging portion, a driving transfer surface **214b** of the pack shutter portion **214** engages with the mounting portion **201** (rotating member) in the pack shutter portion **214**, and as the open/close engaging portion, the driving transfer surface **214b** engages with the main body shutter portion **206** in the pack shutter portion **214**. The pack shutter portion **214** is moved (rotated) together with the mounting portion **201** by the control (operation) force of the mounting portion **201**, and transfers this control force to the main body shutter portion **206**, so as to move the main body shutter portion **206** as well.

(86) In other words, as the force-receiving region, the driving transfer surface **214b** has a region that engages and contacts with the mounting portion driving transfer rib **201d** (rotating member engaged portion). Specifically, as the shutter engaging portions, the driving transfer surface **214b** includes: a first shutter engaging portion **218b1** that is a part of a first groove side surface **214b1** (first recessed portion side surface), out of the pair of groove side surfaces of the groove-shaped portion; and a second shutter engaging portion **218b2** that is a part of a second groove side surface **214b2** (second recessed portion side surface) thereof. On the other hand, as the rotating member

engaged portion, the mounting portion driving transfer rib **201d** includes: a first engaging portion **201dl** (first rotating member engaged portion) that engages with the portion **218b1** of the driving transfer surface **214b** in one circumferential direction around the rotation axial line; and a second engaging portion **201d2** (second rotating member engaged portion) that engages with the portion **218b2** of the driving transfer surface **214b** in the other circumferential direction. Each engaging portion **201dl** and **201d2** of the mounting portion driving transfer rib **201d** has a protruded shape that extends from the inner peripheral surface of the annular mounting portion **201**, centering the rotation axial line RA, toward the rotation axial line RA side.

(87) Furthermore, at force applying regions (a pair of force applying portions), the driving transfer surfaces **214b** include a region that engages and contacts with the main body shutter portion driving transfer rib **206a** (open/close engaged portion). Specifically, as the open/close engaging portion, the driving transfer surfaces **214b** include a portion (first open/close engaging portion) **219b1** provided at a position (downstream side in the inserting direction), which is different from the first shutter engaging portion **218b1** on the first groove side surface in the rotation axial line RA direction, and a portion (second open/close engaging portion) **219b2** provided at a position (downstream side in the inserting direction) which is different from the second shutter engaging portion **218b2** of the second groove side surface in the rotation axial line RA direction. On the other hand, as the open/close engaged portion, the main body shutter portion driving transfer rib **206a** includes a first shutter protruded portion side surface **206al** which is a portion (first open/close engaged portion) that engages with the portion **219b1** of the driving transfer surfaces **214b** in one circumferential direction, and a second shutter protruded portion side surface **206b2** which is a portion (second open/close engaged portion) that engages with the portion **219b2** of the driving transfer surfaces **214b** in the other circumferential direction.

(88) Description of Lever Lock Mechanism (Rotation Control Mechanism)

(89) The lever lock mechanism will be described with reference to FIGS. **24**, **25A**, **25B**, **26A**, and **26B**. The rotation direction of the mounting portion **201** is restricted by the lever lock mechanism, so as to be at the initial position when the replenishing pack **210** is not attached. FIG. **24** is a perspective view depicting the rotation restricting mechanism of the present embodiment, where only the lever lock member **227** (restricting member), the lever lock release link **229** (release member) and the lever lock portion pressing spring **228** (biasing member of the restricting return portion) are illustrated. FIGS. **25A** and **25B** are a bottom view and a cross-sectional view of the mounting portion **201** and the lever lock mechanism when the lever lock member **227** is at the restricting position (locked state). FIGS. **26A** and **26B** are a bottom view and a cross-sectional view of the mounting portion **201** and the lever lock mechanism when the lever lock member **227** is at an allowable position (lock released state).

(90) As illustrated in FIGS. **25A** and **25B**, when the lever lock mechanism is activated, the lever lock member **227** is pressed to the mounting portion **201** by the lever lock portion pressing spring **228**, and the lever lock pressing portion **227a** is inserted into the opening **201e** of the mounting portion **201**. If the mounting portion **201** is rotated in this state, the inner wall of the opening **201e** rotates along the rotation loci R1 and R2, as illustrated in FIG. **25A**. P1 indicates a switching portion where a rotation restricting surface **227b** changes an inclined surface **227d**, and P2 indicates a switching portion at a rotation restricting surface **227c**. When the lever lock mechanism is activated, P1 and P2 are located at the inside of the rotation loci R1 and R2. Therefore the inner wall of the opening **201e** (restricted portion) contacts and engages with the rotation restricting surfaces **227b** and **227c** (restricting portions) which are surfaces perpendicular to the rotating direction, whereby rotation of the mounting portion **201** is restricted.

(91) As illustrated in FIGS. **26A** and **26B**, when the replenishing pack **210** is attached, the lever lock release rib **214a** passes through the lock release recessed portion **201f** and contacts with the lever lock release link **229**. The lever lock release link **229** has an inclined surface shape **229a** that contacts with the lever lock release rib **214a** and functions as a contact activating portion. The lever

lock release link **229** receives a force to move the lever lock member **227** to a pressing position (a position that is moved to the lever lock release position) from the pack shutter portion **214** via the inclined surface shape **229a** and the lever lock release rib **214a**. The inclined surface shape **229a** has a shape that is inclined toward the direction where the replenishing pack **210** is inserted into the mounting portion **201**. In other words, the inclined surface shape **229a** is a shape that is inclined so that the force received from the lever lock release rib **214a** includes the component force that acts in the direction of moving the lever lock release link **229** to the pressing position.

(92) As illustrated in FIG. **26B**, the lever lock release link **229** is pressed in the arrow direction by the force that the inclined surface shape **229a** (pressed portion) receives from the lever lock release rib **214a**, and the edge, where the inclined surface shape **229a** is formed, rises over the side surface of the lever lock release rib **214a**. Thereby the lever lock release link **229** moves to the pressing position where the contact surface **229b** (pressing portion) moves the lever lock member **227** to the lever lock release position. Then the lever lock member **227**, which is in contact with the lever lock release link **229**, moves to the lever lock release position in the arrow direction against the biasing force of the lever lock portion pressing spring **228**.

(93) As illustrated in FIG. **26A**, at the lever lock release position, P1 moves to the outside of the rotation locus R1, and P2 is maintained at the inside of R2. Therefore when the lever portion **201b** is rotated in the arrow direction, the inclined surface **227d** contacts with the inner wall of the opening **201e**, and the lever lock member **227** is retracted by the component force, and the inclined surface **227d** rises over the inner wall of the opening **201e**. Therefore the lever lock member **227** allows rotation of the lever portion **201b** in the arrow direction, and the lever portion **201b** can be rotated to the replenishing position.

(94) The rotation in the opposite direction is continuously restricted since P2 is maintained at the inside of R2. In the present embodiment, the assembling phase of the mounting portion **201** is set to the direction in which rotation is restricted, whereby both the prevention of the rotation to the assembling phase direction after assembly and the rotation up to the replenishing position are implemented.

(95) FIGS. **27A** to **27E** indicate a flow from the initial position of the lock mechanism to the replenishing position. In the initial position illustrated in FIG. **27A**, P1 and P2 are on the inside of the rotation loci R1 and R2, and the rotation of the mounting portion **201** is restricted thereby, as described above. As illustrated in FIG. **27B**, when the replenishing pack **210** is attached, the lever lock member **227** is retracted, and the mounting portion **201** can be rotated in the arrow direction, as described above. FIG. **27C** indicates the state where the lock member **227** is retracted, and the inclined surface **227d** completely rises over the opening **201e**. From this state to the replenishing position, the lock member **227** and the lever lock release link **229** are in relative positions, and do not contact with each other (separated state), as illustrated in FIG. **27D**, and the lock member **227** contacts with the outer surface of the mounting portion **201**, and is maintained in the rotation allowable position. Since the lock member **227** does not interfere with the rotation of the mounting portion **201** in this state, the user can move the lever portion **201b** to the replenishing position. If the mounting portion **201** is rotated to the replenishing position, as illustrated in FIG. **27E**, the lever lock pressing portion **227a** enters the replenishing position recessed portion **201h** of the mounting portion **201**, where an audible sound or operation force changes. Thereby, the user can easily recognize that the lever portion **201b** has been move to the replenishing position.

(96) Replenishing Container Mounting Procedure

(97) The procedure of replenishing toner using the replenishing pack **210** will be described with reference to FIGS. **28A**, **28B**, **29A**, and **29B**. FIGS. **28A** and **28B** are perspective views of the image forming apparatus **1**. FIG. **28A** indicates a state in which the discharge tray **14** covers the attached portion **200**, and is at a position where the recording material P, discharged from the discharge outlet **15**, can be loaded. FIG. **28B** indicates a state in which the discharge tray **14** is at a position where the attached portion **200** is exposed from the discharge tray **14**. The discharge tray

14 is configured to be movable to the position where the recording material **P** can be loaded, as indicated in FIG. **28A**, and the position where the attached portion **200** is exposed, as indicated in FIG. **28B**. The attached portion **200** is provided at an upper part of the front surface of the image forming apparatus **1**, hence the user has easy access to the portion when toner is replenished.

(98) When the toner is replenished, the recording material **P**, loaded on the discharge tray **14**, is removed, and the discharge tray **14** at the loadable position indicated in FIG. **28A**, is opened to be moved to the expose position indicated in FIG. **28B**. When the discharge tray **14** is opened, the attached portion **200** and a top surface portion **240**, which adjoins the attached portion **200**, is exposed. Then the replenishing pack **210** is inserted into the exposed attached portion **200**. At this time, the replenishing pack **210** is inserted such that the position of the mounting portion driving transfer rib **201d** (FIG. **19B**), which is provided in the attached portion **200**, matches with the position of the positioning portion **217** (FIG. **18B**) provided in the replenishing pack **210**. If the positions of the mounting portion driving transfer rib **201d** and the positioning portion **217** do not match, the replenishing pack **210** interferes with the mounting portion driving transfer rib **201d**, and the replenishing pack **210** cannot be attached.

(99) Here, if the colors of the mounting portion **201** and the lever lock release link **229** are different, then the phases of the replenishing pack **210** and the mounting portion **201** can, in a visual sense, be easily matched when the replenishing pack **210** is attached.

(100) FIG. **29A** indicates a state where the replenishing pack **210** is inserted into the attached portion **200**. As illustrated in FIG. **29A**, in the present embodiment, the replenishing pack **210** can be inserted when the direction **D**, in which the pouch edge **216** extends, is parallel with the **X** direction. When the replenishing pack **210** is inserted into the base of the attached portion **200**, the main body shutter portion driving transfer rib **206a** (FIG. **14**) of the attached portion **200** engages with the positioning portion **217** (FIGS. **18A** and **18B**) of the pack shutter portion **214** of the replenishing pack **210**. In other words, the rib **206a** faces the positioning portion **217** in the radius direction, and is inserted between the driving transfer surfaces **214b**, whereby the main body shutter portion **206** and the pack shutter portion **214** are engaged and integrated. As a result, the main body shutter portion **206** and the pack shutter portion **214** can be rotated in tandem by the operation of the lever portion **201b**.

(101) FIG. **29B** indicates a state where the lever portion **201b** is moved from the initial position to the replenishing position. At this time, the replenishing pack **210** in the **Z** direction is fixed to the attached portion **200** by a replenishing pack retaining mechanism (not illustrated). Then, as mentioned above, the pack shutter portion **214**, provided in the replenishing pack **210**, is rotated by moving the lever portion **201b**. Further, the main body shutter portion **206** of the attached portion **200** is rotated with the pack shutter portion **214** since the main body shutter portion **206** and the pack shutter portion **214** are engaged. As a result, when the lever portion **201b** is moved to the replenishing position, the side surface opening **205** (FIGS. **15A** to **15C**), which is formed in the toner-receiving portion **202**, opens, and at the same time, the opening **213** (FIGS. **18A** and **18B**), which is formed in the inserting portion **212**, opens. The side surface opening **205** formed in the toner-receiving portion **202** and the opening **213** formed in the inserting portion **212** are in such a positional relationship that the side surface opening **205** and the opening **213** face each other at the point when the replenishing pack **210** is inserted into the attached portion **200**. Therefore when the lever portion **201b** is moved from the initial position to the replenishing position, the replenishing pack **210**, the attached portion **200** and the container portion **18** are connected, and the toner can be replenished.

(102) FIG. **30** indicates a top view when the state in FIG. **29B** is viewed from the top. In the replenishing pack **210** attached to the image forming apparatus **1** illustrated here, the direction **D** extending from the pouch edge **216** is parallel with the **X** direction, as mentioned above. At the edge on the plus side of the **Y** direction (front surface side) of the top surface portion **240** which is exposed when the discharge tray **14** is opened, a protruded portion **241**, which protrudes in the plus

side of the Z direction (upper side), is formed. The notch **242** is formed in a part of the protruded portion **241**, and the position where the notch **242** is formed corresponds to the rotation locus of the lever portion **201b**.

(103) In FIG. **30**, the lever portion **201b** at the initial position is indicated by a dotted line. After the toner replenishment completes, the lever portion **201b** is returned to the original position (initial position). At this time, in reverse sequence of the lever operation to the replenishing position, both the main body shutter portion **206** of the attached portion **200** and the pack shutter portion **214** of the replenishing pack **210** rotate first, then both the side surface opening **205** and the opening **213** thereof close respectively. Then the retention of the attached portion **200** and the replenishing pack **210** is released, and the replenishing pack **210** can be detached from the attached portion **200**. Thereby the pack shutter portion **214** is in the closed state unless the replenishing pack **210** is inserted into the attached portion **200** of the image forming apparatus **1**, and leaking of toner can be prevented.

Embodiment 2

(104) In the configuration of using the replenishing pack **210**, the user attaches the replenishing pack **210**, and replenishes the toner to the image forming apparatus **1**. Therefore if the user attaches a replenishing pack **210** filled with incompatible toner in error, image problems may occur. In Embodiment 1, the lock release recessed portion **201f** of the mounting portion **201** used for the lever lock mechanism, is formed to be larger compared with the lock release rib **214a** of the replenishing pack **210**, and the lever lock mechanism can be released even if the positions of the lock release recessed portion **201f** and the lock release rib **214a** deviate somewhat. Therefore in Embodiment 2, as illustrated in FIGS. **31A** and **31B**, the positional relationship of the lock release recessed portion **201f** and the lock release rib **214a** is designed such that only a compatible combination is allowed, so that an attachment error by the user can be prevented.

(105) FIG. **31A** indicates a first replenishing pack **210A** with which the lock release recessed portion **201f** and the lock release rib **214a** do not interfere with each other. In other words, the first replenishing pack **210A** is at such a relative position that various guide portions of the first replenishing pack **210A** can be engaged with the attached portion **200**, and a lock release rib **214aA** can be guided by a lock release recessed portion **201f** (guide portion). FIG. **31B**, on the other hand, indicates a second replenishing pack **210B** with which a lock release rib **214aB** is provided at a position that is different from the lock release rib **214aA**. The engaging positions of various guide portions of the second replenishing pack **210B** are the same as the first replenishing pack **201A**, but the position of the lock release rib **214aB** is at a position where engaging with the lock release recessed portion **201f** is not allowed, hence the second replenishing pack **210B** contacts with the mounting portion driving transfer rib **201d**, which is adjacent to the lock release recessed portion **201f**, and forms a wall portion (attachment restricting portion). Thereby the second replenishing pack **210B** is restricted not to be attached to the attached portion **200**. The phases of the replenishing container **210** and the mounting portion **201**, when the replenishing pack **210** is attached, are determined by the engagement of the positioning portion **217** and the mounting portion driving transfer rib **201d**, hence if the position of the lock release rib **214a** is different, the lock release rib **214a** interferes with the mounting portion driving transfer rib **201d**, whereby the replenishing pack **210** cannot be completely attached. Therefore by disposing a lock release rib **214a** at a different position in the replenishing pack **210** filled with incompatible toner, an attachment error by the user can be prevented.

(106) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(107) This application claims the benefit of Japanese Patent Application No. 2020-218436, filed on Dec. 28, 2020, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image forming apparatus to which a developer container is attachable, the developer container provided with a discharge port and a container shutter configured to move for opening/closing the discharge port, the image forming apparatus comprising: (i) a developer accommodating portion; and (ii) an attached portion to which the developer container is to be attached in an attaching direction and provided with a receiving port through which the developer accommodating portion is to receive developer from the developer container, the attached portion including, (ii-i) an operation member configured to move and configured to be operated by a user to move the container shutter, the operation member having an engaging portion configured to engage with the container shutter in a case where the developer container is attached to the attached portion; (ii-ii) a restriction member configured to move, in a direction intersecting with the attaching direction, between a restricting position in which the restriction member restricts movement of the operation member by engaging with the operation member and an allowable position in which the restriction member allows the movement of the operation member, the restriction member being in the restricting position in a case where the developer container is not attached to the attached portion, wherein the restriction member is configured to move from the restricting position to the allowable position by an attachment of the developer container to the attached portion, and (ii-iii) a biasing member that biases the restriction member in a direction in which the restriction member is moved from the allowable position to the restricting position.
2. The image forming apparatus according to claim 1, wherein the attached portion further includes: a release member that releases restriction of the movement of the operation member, the release member including a pressed portion and a pressing portion to press the restriction member in a direction in which the restriction member is moved from the restricting position to the allowable position; wherein the release member is moved by the pressed portion being pressed by the developer container in the case where the developer container is attached to the attached portion, so that the pressing portion of the release member presses the restriction member to move the restriction member from the restricting position to the allowable position against a biasing force of the biasing member.
3. The image forming apparatus according to claim 2, wherein the release member is movably supported by the operation member so as to move together with the operation member, and wherein the attached portion includes a frame by which the restriction member is movably supported and to which the biasing member is attached.
4. The image forming apparatus according to claim 1, wherein when the restriction member is in the restricting position, the restriction member restricts the operation member from being moved from a first position to a second position, and when the restriction member is in the allowable position, the restriction member allows the operation member to be moved between the first position and the second position, and wherein in a state that the developer container is attached to the attached portion and the operation member is in the first position, the developer accommodating portion is not capable of receiving the developer from the developer container, and in a state that the developer container is attached to the attached portion and the operation member is in the second position, the developer accommodating portion is capable of receiving the developer from the developer container.
5. The image forming apparatus according to claim 4, wherein the attached portion includes an apparatus shutter which is provided with an apparatus shutter opening and a shutter engaging portion configured to engage with the container shutter and which is configured to move between a communication position where the apparatus shutter opening is communicated with the receiving port and a non-communication position where the apparatus shutter opening is not communicated with the receiving port, wherein the operation member is configured to move, through the container

shutter, the apparatus shutter between the communication position and the non-communication position in a case where the developer container is attached to the attached portion, and wherein when the operation member is in the first position, the apparatus shutter is in the non-communication position, and when the operation member is in the second position, the apparatus shutter is in the communication position.

6. An image forming apparatus to which a developer container is attachable, the image forming apparatus comprising: (i) a developer accommodating portion; and (ii) an attached portion to which the developer container is to be attached in an attaching direction and provided with a receiving port through which the developer accommodating portion is to receive developer from the developer container, the attached portion including, (ii-i) an apparatus shutter which is provided with an opening and which is configured to move between a communication position where the opening is communicated with the receiving port and a non-communication position where the opening is not communicated with the receiving port, (ii-ii) an operation member configured to move and configured to be operated by a user for movement of the apparatus shutter between the communication position and the non-communication position; (ii-iii) a restriction member movable, in a direction intersecting with the attaching direction, between a restricting position in which the restriction member restricts movement of the operation member by engaging with the operation member and an allowable position in which the restriction member allows the movement of the operation member for moving the apparatus shutter between the communication position and the non-communication position, the restriction member being in the restricting position in a case where the developer container is not attached to the attached portion, the restriction member being configured to move from the restricting position to the allowable position by an attachment of the developer container to the attached portion; and (ii-iv) a biasing member that biases the restriction member in a direction in which the restriction member is moved from the allowable position to the restricting position.

7. The image forming apparatus according to claim 6, wherein the attached portion further includes: a release member that releases restriction of the movement of the operation member, the release member including a pressed portion, and a pressing portion to press the restriction member in a direction in which the restriction member is moved from the restricting position to the allowable position; wherein the release member is moved by the pressed portion being pressed by the developer container in the case where the developer container is attached to the attached portion, so that the pressing portion of the release member presses the restriction member to move the restriction member from the restricting position to the allowable position against the biasing force of a biasing member.

8. The image forming apparatus according to claim 7, wherein the release member is movably supported by the operation member so as to move together with the operation member, and wherein the attached portion includes a frame by which the restriction member is movably supported and to which the biasing member is attached.

9. An image forming apparatus to which a developer container is attachable, the image forming apparatus comprising: (i) a developer accommodating portion; and (ii) an attached portion to which the developer container is to be attached in an attaching direction and provided with a receiving port through which the developer accommodating portion is to receive developer from the developer container, the attached portion including, (ii-i) an apparatus shutter which is provided with an opening and which is configured to move between a communication position where the opening is communicated with the receiving port and a non-communication position where the opening is not communicated with the receiving port, (ii-ii) a restriction member movable, in a direction intersecting with the attaching direction, between a restricting position in which the restriction member restricts movement of the apparatus shutter between the communication position and the non-communication position and an allowable position in which the restriction member allows the movement of the apparatus shutter, the restriction member being in the

restricting position in a case where the developer container is not attached to the attached portion, the restriction member being configured to move from the restricting position to the allowable position by an attachment of the developer container to the attached portion, and (ii-iii) a biasing member that biases the restriction member in a direction in which the restriction member is moved from the allowable position to the restricting position.
